

From Slides to Space: Interactive Scale Navigation for XR Presentations

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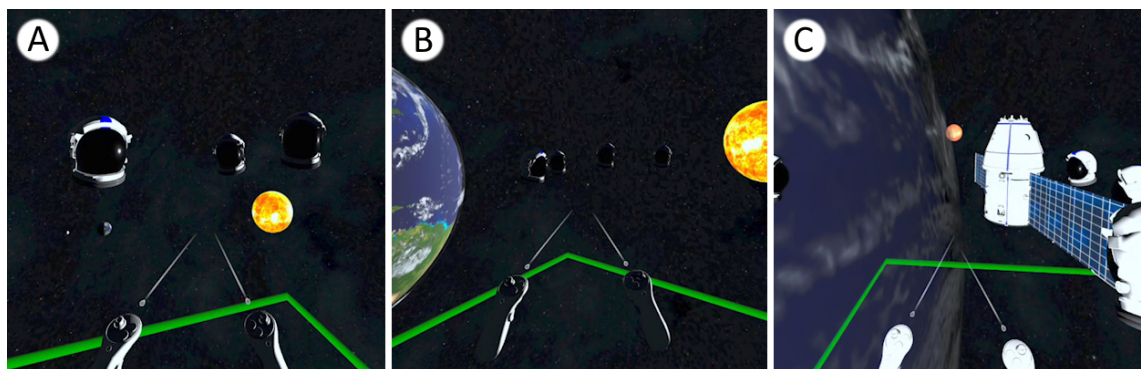


Figure 1: Scale-based translation across regions: (A) Solar system scale (B) Earth Scale. (C) Satellite Scale.

ABSTRACT

This paper presents our solution to the 2026 3DUI Contest challenge. Our approach features scale-based navigation and spatialized regions to traverse an immersive presentation, replacing traditional slide decks with embodied movement through a 3D virtual environment (VE). To demonstrate this approach, we designed an immersive presentation centered on the exploration of the solar system, where predetermined regions function as presentation slides. The user, acting as the presenter, exhibits content by scaling and moving the VE over the course of the tour, allowing for both large-scale context and close inspection of details while maintaining clear instructional flow. The system integrates presenter-oriented tools commonly used in conventional presentations along with visual and auditory feedback from audience participation. The guided tour provides narrative segments comparable to a presentation slide while allowing attendees to explore current and previously visited regions. This work explores how virtual reality (VR) can extend conventional presentation methods through spatial interaction and embodied navigation.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality ; Human-centered computing—Human computer interaction (HCI)—Interaction techniques

1 INTRODUCTION

Traditional slide-based presentations rely on 2D linear sequencing, which can limit engagement and make it difficult to convey complex concepts [1]. The two primary limitations we seek to address are the abstract nature of slides and the passive experience of fixed-point observation, both of which stem from the 2D linear format of traditional presentations. In contrast, VR enables embodied engagement with information in a virtual environment (VE),

supporting learning through interaction, movement, and spatial exploration [2]. In this paper, we introduce an immersive presentation approach in which content progression is driven by scale and spatial navigation rather than traditionally discrete slide transitions. Through the use of a guided tour, our system integrates narrative structure, presenter-oriented interaction tools, and multi-modal audience feedback to support instructional clarity and exploratory engagement [3].

2 DESIGN PROCESS

Our design process began with weekly meetings to analyze and compare existing presentation modalities, tools, and limitations, and to assess their applicability to immersive experiences. We explored how commonly used systems have solved many of the issues of 2D presentations and identified shortcomings that VR could overcome. The team followed an iterative workflow, holding weekly meetings to discuss high-level design, review progress, and coordinate tasks. Optional “office hours” were held weekly to provide team members with a dedicated time and space to complete assigned tasks. Task assignment, progress tracking, and design notes were maintained using Trello¹.

Prototypes were developed and evaluated using various Meta Quest 3 head-worn displays and corresponding controllers. The experience was implemented in the Unity Game Engine² using the OpenXR framework to ensure compatibility across hardware.

3 NARRATIVE

To ground our exploration of immersive presentations, we developed a narrative scenario that motivated our design. The scenario takes place in a future where museums have all but been lost to the overwhelming demand for public space and curators have been compelled to transform their exhibits into remote experiences. Over time, they recognized that traditional presentation methods do not engage classroom audiences as effectively as in-person exhibits. Instead, they have sought to develop software that can bring museums virtually to the classroom. Our story centers on the user

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¹<https://trello.com/>

²<https://unity.com/>

who assumes the role of a planetarium presenter, guiding a class of students through an educational tour of the solar system, with a primary focus on the formation and impact of sunspots on Earth and our various satellites. The tour progresses around various astronomical objects within the solar system, allowing the presenter to navigate through predetermined scaled regions of interest.

4 APPROACH AND INTERACTIONS

This section describes the primary interaction and navigation mechanisms of the system.

4.1 Spatial Regions as Replacement for Slides

In lieu of a traditional slide deck, our approach structures the presentation as a series of predefined regions within the 3D space. Each region represents a specific topic of discussion for the presenter and is positioned and scaled to best correspond to the physical representation of the discussion topic. Within these regions, attendees are free to roam within the defined boundaries and explore concepts while remaining engaged with and close to the presenter.

4.2 Space-Based Navigation through Regions

At designated moments, the audience can be released from direct guidance and allowed to explore freely. During these periods, they are encouraged to explore the surrounding content, moving through both space and scale. They can also use a hand-raise interface to signal the presenter, indicating their location with both a visual highlight and auditory ping. Although the perspectives of the attendees were considered, the demonstration is limited to the functionality and toolset of the presenter due to the foreseen complexity of multiplayer development.

4.3 Presenter Toolset

To support the presenter throughout the tour, we integrated a suite of presenter-oriented tools into the VE. The presenter's dominant hand features a circular toolbar menu containing traditional presentation tools as seen in Fig. 2A, including highlighting, flagging, pointing, and drawing mechanisms for directing attention. These tools mirror familiar presentation affordances while leveraging the spatial and embodied nature of VR, facilitating adoption by aligning with established interaction paradigms [4].

The drawing tool enables annotation directly within the environment, allowing the presenter to reinforce key concepts at multiple spatial scales. The tool includes an additional action button that toggles a fade-out effect to not permanently clutter the VE.

The laser pointer, flagging tool, and highlighter each provide distinct forms of directional guidance during explanation. The flagging tool and laser pointer anchor the audience's attention to important features within the VE while the highlighter directs focus toward specific objects.

In addition, an embedded mini-map, attendee summary, and speaker notes panel are accessible from the presenter's nondominant hand when raised as seen in Fig. 2B. Together, these elements allow the tour to function as a fully realized replacement for traditional presentation techniques in an immersive VE.

5 4-DOF VIEWPOINT CONTROL

The presenter is able to scale the environment in two distinct ways: by progressing through regions constructed prior to presenting, and by scaling the environment in real time. By holding both grip buttons and moving their hands apart or together, the presenter can scale the VE along with the audience, allowing for granular inspection of fine details as well as a large-scale overview.[5] Additionally, the presenter can move the environment around themselves by holding down a single grip button and moving their hand should they wish to encourage the group to take a closer look at some key detail that may otherwise be overlooked.

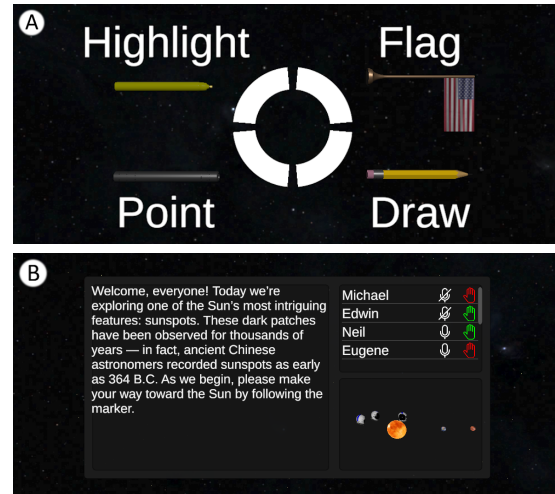


Figure 2: The tool menus for interactions: (A) The dominant hand menu featuring a circular toolbar menu. (B) The nondominant hand menu featuring the map, attendee summary, and speaker notes.

When the presenter moves between regions or scales the space within a region, the audience is physically carried along with the evolving environment while retaining freedom to explore locally, supporting both shared progression and individual sense-making. To reduce potential discomfort from forced movement, users and attendees can adjust their comfort settings to either apply a vignette or temporarily blackout the view during transitions. In addition, all scale-based movements were made as smooth as possible to mitigate motion sickness during aggressive perspective changes. Research in embodied learning suggests that movement through space can aid cognitive processing and retention by coupling physical activity with conceptual understanding [6]. As a result, scale transitions function as an organizational mechanism for structuring instructional content, providing clear contextual shifts while maintaining spatial continuity.

6 CONCLUSION

Our system demonstrates that replacing slide-based workflows with spatial navigation in VR can significantly improve communication of complex material by reducing abstraction, increasing engagement, and strengthening conceptual understanding. The solar system tour shows that spatial scale and movement serve as powerful organizational mechanisms, enabling deeper comprehension and more effective knowledge transfer than traditional slides.

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